UNIVERSIDAD NACIONAL DE SAN AGUSTÍN

FACULTAD DE INGENIERÍA DE PRODUCCIÓN Y SERVICIOS

ESCUELA PROFESIONAL DE CIENCIA DE LA COMPUTACIÓN

BIG DATA

Ejercicios de Hive

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AREQUIPA – PERÚ

2025

EJERCICIO 1: WORDCOUNT

```sql
CREATE TABLE docs (line STRING);

LOAD DATA LOCAL INPATH '/home/hivehadoop/wordcount.txt'
OVERWRITE INTO TABLE docs;

CREATE TABLE word_count AS

SELECT word, count(1) AS count

FROM (

 SELECT explode(split(line, ' ')) AS word

 FROM docs

) w

GROUP BY word

ORDER BY word;
```

1. CREATE TABLE docs (line STRING);`

Hive crea una tabla administrada llamada docs.

- En el sistema de archivos (por defecto HDFS), se crea una carpeta: /user/hive/warehouse/docs
- Esta tabla tiene una sola columna de tipo STRING, que almacenará cada línea completa del archivo como un solo registro.
- Se registra en el Metastore (la base de datos interna de Hive que almacena metadatos de tablas).
- No se cargan datos aún, solo se crea la estructura.

2. LOAD DATA LOCAL INPATH ... INTO TABLE docs

```sql
LOAD DATA LOCAL INPATH '/home/hivehadoop/wordcount.txt'
OVERWRITE INTO TABLE docs;
```

Carga el contenido del archivo wordcount.txt desde el sistema de archivos local (no desde HDFS).

- Cada línea del archivo se convierte en una fila en la tabla docs.
- La opción OVERWRITE borra cualquier dato anterior en la tabla.
- Hive copia el archivo desde /home/hivehadoop/wordcount.txt hacia: /user/hive/warehouse/docs/ en HDFS.
- No interpreta el contenido aún, solo lo almacena como está (líneas de texto).
- Usa el InputFormat predeterminado TextInputFormat, que considera cada línea como un registro.

3. CREATE TABLE word_count AS SELECT ...

```SQL
CREATE TABLE word_count AS
SELECT word, count(1) AS count
FROM (
  SELECT explode(split(line, ' ')) AS word
  FROM docs
) w
GROUP BY word
ORDER BY word;
```

¿Qué hace?
Este bloque realiza el procesamiento principal del WordCount. Vamos paso por paso:

a) split(line, ' ')

Divide el contenido de cada línea usando espacios como separador.

Resultado: un array de palabras por línea.
Ejemplo:

```SQL
split('Hola mundo hola', ' ') → ['Hola', 'mundo', 'hola']
```

b) explode(...)

Toma cada array y genera una fila por cada elemento del array.

Ejemplo:

```SQL
explode(['Hola', 'mundo', 'hola']) →

  'Hola'

  'mundo'

  'hola'
```

Así, de una línea se generan múltiples registros.

c) FROM docs

Hive recorre cada fila de la tabla docs y aplica split() y explode() a cada una.

Resultado: una tabla temporal con muchas filas, una por cada palabra.

d) GROUP BY word

Agrupa todas las filas por la palabra (word), es decir, pone juntas todas las ocurrencias de cada palabra.

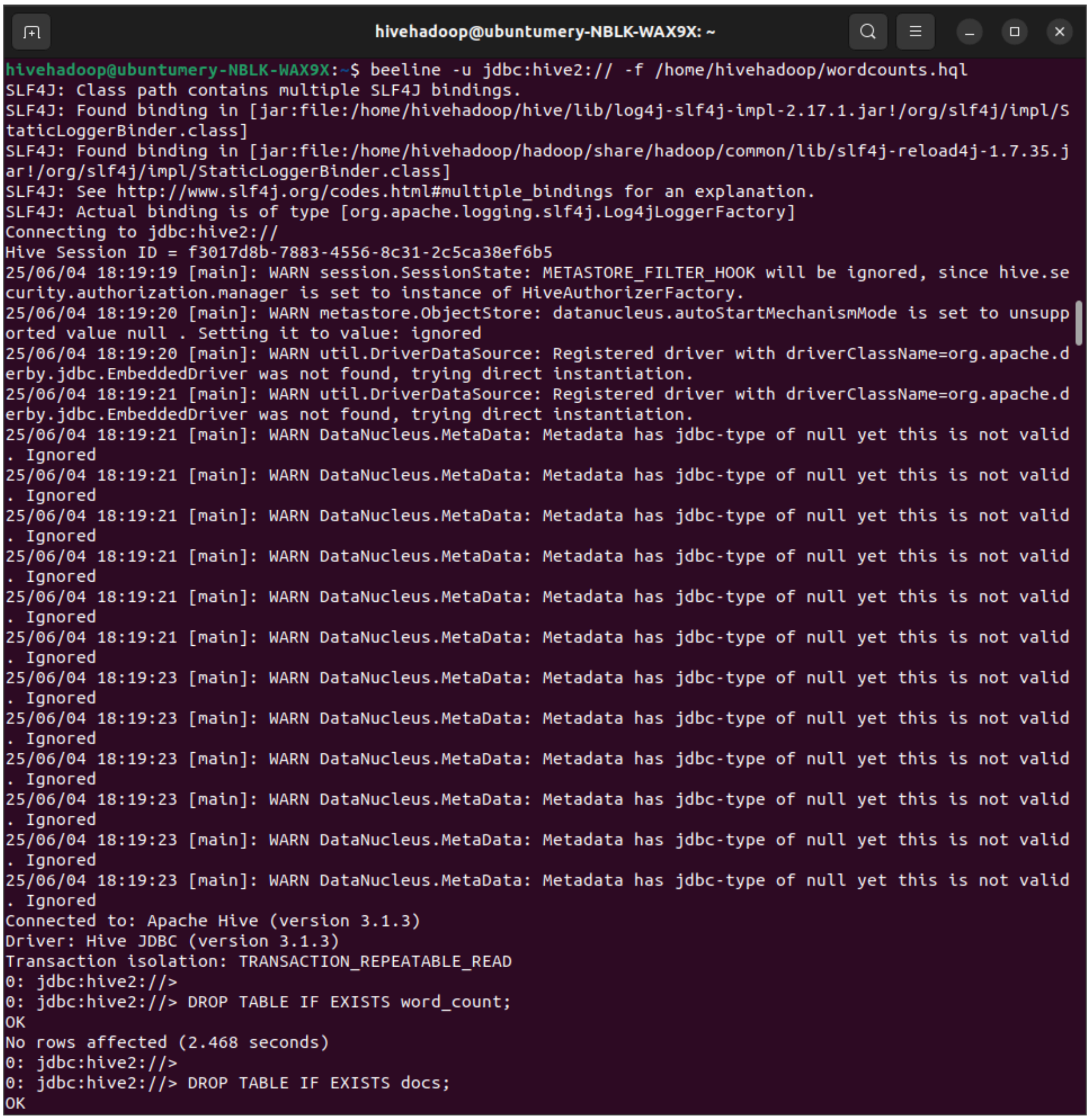
e) count(1) AS count

Cuenta cuántas veces aparece cada palabra.
count(1) es una forma eficiente de contar filas por grupo (no depende de contenido nulo).

f) CREATE TABLE word_count AS (...)

Hive almacena el resultado de la consulta como una nueva tabla física llamada word_count.

- Crea la carpeta correspondiente:
  /user/hive/warehouse/word_count
- Inserta los resultados como archivos tipo texto plano.



```
hivehadoop@ubuntumery-NBLK-WAX9X: ~
hivehadoop@ubuntumery-NBLK-WAX9X:~$ beeline -u jdbc:hive2:// -f /home/hivehadoop/wordcounts.hql
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/home/hivehadoop/hive/lib/log4j-slf4j-impl-2.17.1.jar!/org/slf4j/impl/S
taticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/hivehadoop/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.35.j
ar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Connecting to jdbc:hive2://
Hive Session ID = f3017d8b-7883-4556-8c31-2c5ca38ef6b5
25/06/04 18:19:19 [main]: WARN session.SessionState: METASTORE_FILTER_HOOK will be ignored, since hive.se
curity.authorization.manager is set to instance of HiveAuthorizerFactory.
25/06/04 18:19:20 [main]: WARN metastore.ObjectStore: datanucleus.autoStartMechanismMode is set to unsupp
orted value null . Setting it to value: ignored
25/06/04 18:19:20 [main]: WARN util.DriverDataSource: Registered driver with driverClassName=org.apache.d
erby.jdbc.EmbeddedDriver was not found, trying direct instantiation.
25/06/04 18:19:21 [main]: WARN util.DriverDataSource: Registered driver with driverClassName=org.apache.d
erby.jdbc.EmbeddedDriver was not found, trying direct instantiation.
25/06/04 18:19:21 [main]: WARN DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid
. Ignored
25/06/04 18:19:21 [main]: WARN DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid
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25/06/04 18:19:23 [main]: WARN DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid
. Ignored
Connected to: Apache Hive (version 3.1.3)
Driver: Hive JDBC (version 3.1.3)
Transaction isolation: TRANSACTION_REPEATABLE_READ
0: jdbc:hive2://>
0: jdbc:hive2://> DROP TABLE IF EXISTS word_count;
OK
No rows affected (2.468 seconds)
0: jdbc:hive2://>
0: jdbc:hive2://> DROP TABLE IF EXISTS docs;
OK
```

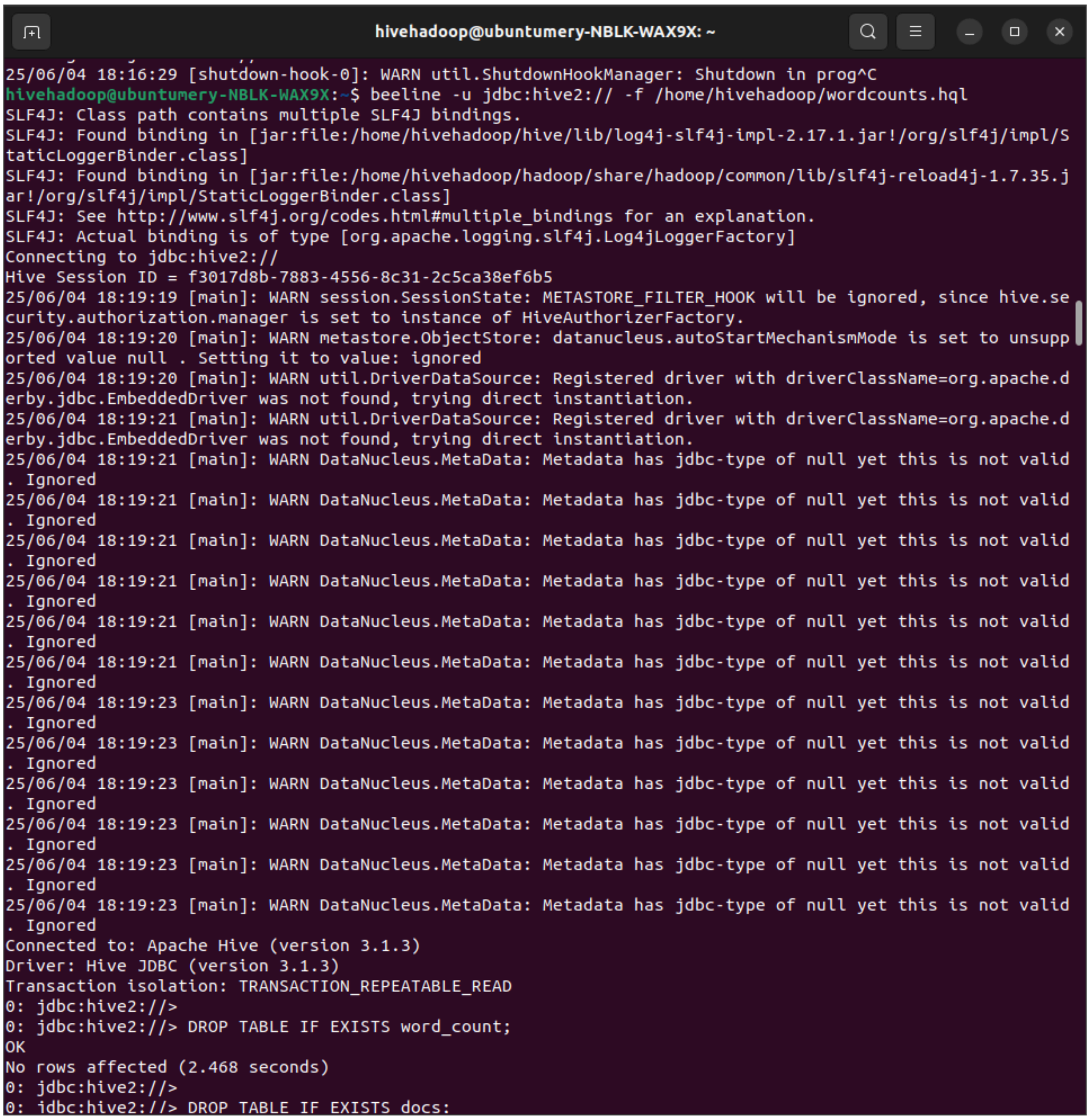

```
hivehadoop@ubuntumery-NBLK-WAX9X: ~
25/06/04 18:16:29 [shutdown-hook-0]: WARN util.ShutdownHookManager: Shutdown in prog^C
hivehadoop@ubuntumery-NBLK-WAX9X:~$ beeline -u jdbc:hive2:// -f /home/hivehadoop/wordcounts.hql
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/home/hivehadoop/hive/lib/log4j-slf4j-impl-2.17.1.jar!/org/slf4j/impl/S
taticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/hivehadoop/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.35.j
ar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Connecting to jdbc:hive2://
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. Ignored
Connected to: Apache Hive (version 3.1.3)
Driver: Hive JDBC (version 3.1.3)
Transaction isolation: TRANSACTION_REPEATABLE_READ
0: jdbc:hive2://>
0: jdbc:hive2://> DROP TABLE IF EXISTS word_count;
OK
No rows affected (2.468 seconds)
0: jdbc:hive2://>
0: jdbc:hive2://> DROP TABLE IF EXISTS docs;
```

EJERCICIO 2: Número de entradas en el log para cada usuario

SQL

```sql
DROP TABLE IF EXISTS logs;

DROP TABLE IF EXISTS result;

CREATE TABLE logs (

  user_a STRING,
```

```
 visit_time STRING,
 query STRING
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY '\t';
LOAD DATA LOCAL INPATH '/home/hivehadoop/excite-small.log'
OVERWRITE INTO TABLE logs;
CREATE TABLE result AS
SELECT user_a, COUNT(1) AS log_entries
FROM logs
GROUP BY user_a
ORDER BY user_a;
SELECT * FROM result;
```

1. DROP TABLE IF EXISTS logs; y DROP TABLE IF EXISTS result;

- Elimina las tablas logs y result si ya existen.
- Esto evita errores en caso de que las ejecuciones anteriores hayan creado estas tablas.

2. CREATE TABLE logs (...)

SQL

```
CREATE TABLE logs (
 user_a STRING,
 visit_time STRING,
 query STRING
)
```

```
ROW FORMAT DELIMITED

FIELDS TERMINATED BY '\t';
```

- Crea una tabla llamada logs con las siguientes columnas:
  - user_a: ID del usuario
  - visit_time: fecha y hora de la búsqueda
  - query: cadena de búsqueda que ingresó el usuario
- ROW FORMAT DELIMITED: indica que se usará un delimitador de campo.
- FIELDS TERMINATED BY el delimitador es un **tabulador** (\t`), típico en archivos de logs como este.

Se crea en HDFS: /user/hive/warehouse/logs

3. LOAD DATA LOCAL INPATH ... INTO TABLE logs

```
SQL

LOAD DATA LOCAL INPATH '/home/hivehadoop/excite-small.log'

OVERWRITE INTO TABLE logs;
```

- Carga los datos del archivo excite-small.log desde el sistema de archivos local a la tabla logs.

Procesos que ocurren:

- Se copia el archivo al HDFS.
- Cada línea se interpreta como un registro.
- Los campos se separan por tabulaciones y se asignan a user_a, visit_time y query.

4. CREATE TABLE result AS SELECT ...

```
SQL

CREATE TABLE result AS

SELECT user_a, COUNT(1) AS log_entries

FROM logs
```

```
GROUP BY user_a

ORDER BY user_a;
```

- Crea una nueva tabla result con el conteo de entradas de logs por cada usuario.

Detalles del proceso:

1. FROM logs: lee todos los registros de la tabla logs.
2. GROUP BY user_a: agrupa las entradas por usuario.
3. COUNT(1): cuenta cuántos registros tiene cada usuario.
4. ORDER BY user_a: ordena alfabéticamente los resultados por ID de usuario.

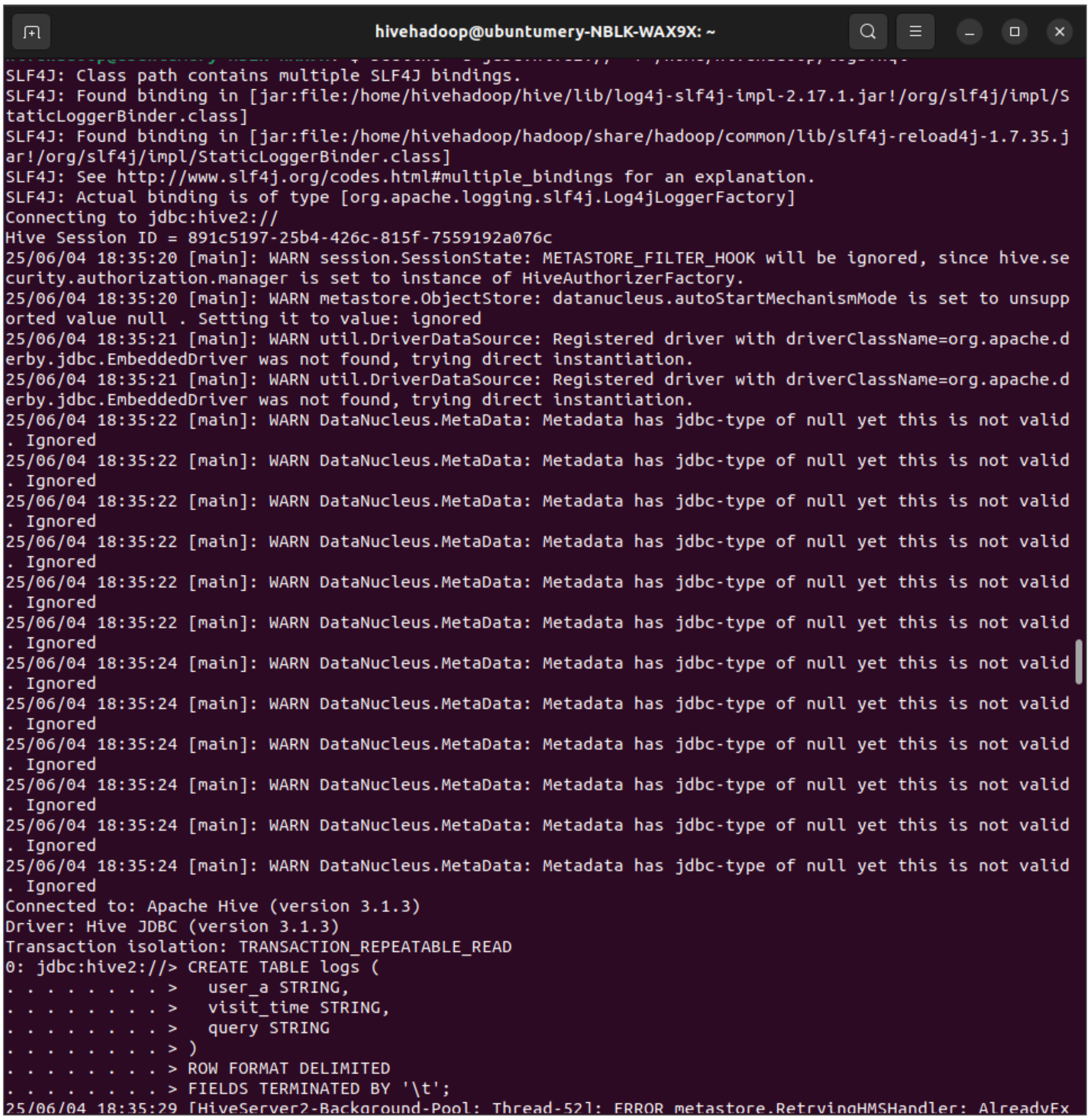
Se crea la tabla física result con dos columnas:

- user_a: identificador del usuario
- log_entries: número total de registros para ese usuario

5. SELECT * FROM result;

- Muestra todos los resultados almacenados en la tabla result.

Resultado esperado: Una tabla con cada user_a seguido del número de veces que aparece en el archivo de logs (es decir, cuántas búsquedas realizó).


hivehadoop@ubuntumery-NBLK-WAX9X: ~
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/home/hivehadoop/hive/lib/log4j-slf4j-impl-2.17.1.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/hivehadoop/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.35.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Connecting to jdbc:hive2://
Hive Session ID = 891c5197-25b4-426c-815f-7559192a076c
25/06/04 18:35:20 [main]: WARN session.SessionState: METASTORE_FILTER_HOOK will be ignored, since hive.security.authorization.manager is set to instance of HiveAuthorizerFactory.
25/06/04 18:35:20 [main]: WARN metastore.ObjectStore: datanucleus.autoStartMechanismMode is set to unsupported value null . Setting it to value: ignored
25/06/04 18:35:21 [main]: WARN util.DriverDataSource: Registered driver with driverClassName=org.apache.derby.jdbc.EmbeddedDriver was not found, trying direct instantiation.
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25/06/04 18:35:22 [main]: WARN DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
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25/06/04 18:35:24 [main]: WARN DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
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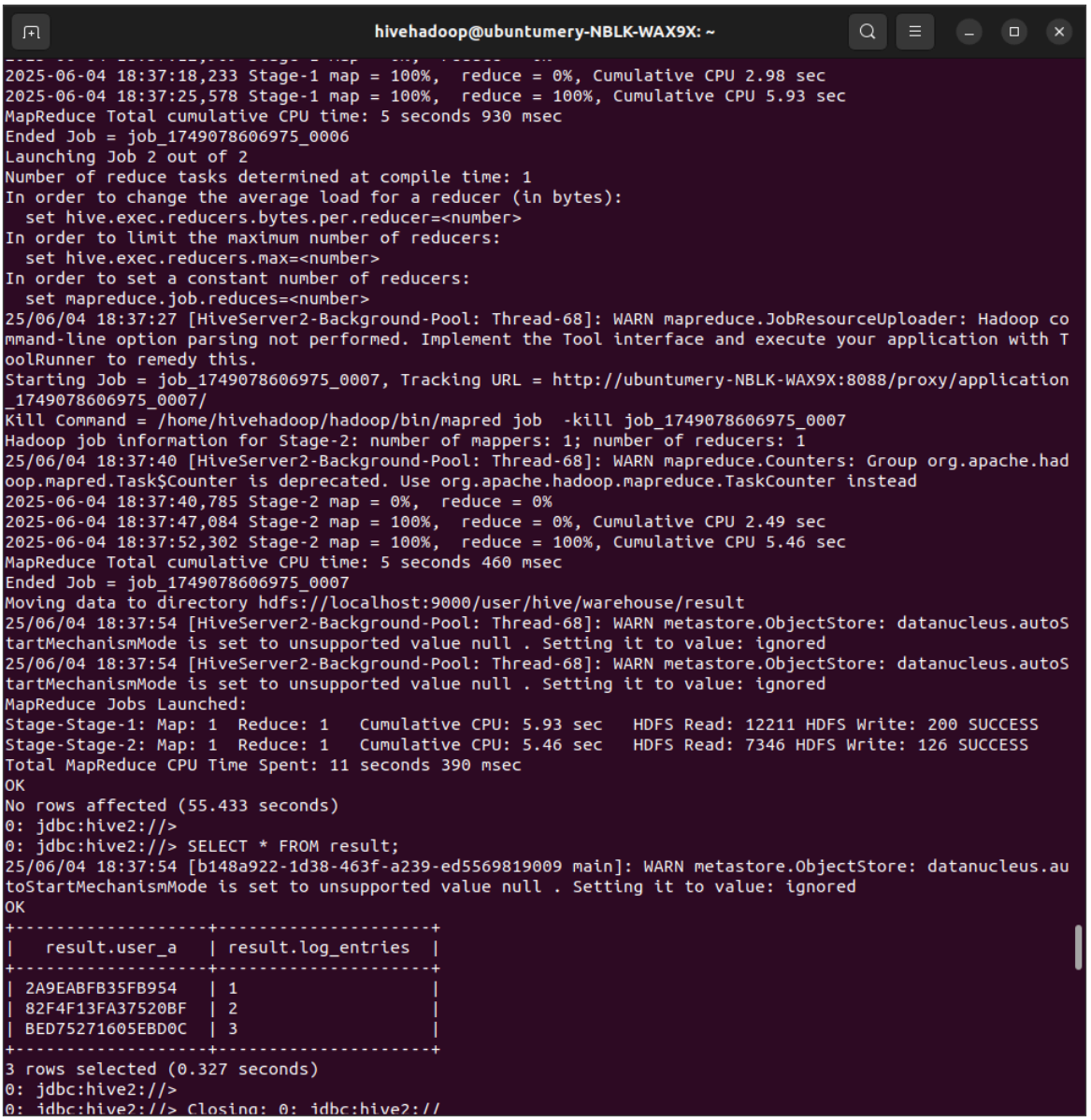
25/06/04 18:35:24 [main]: WARN DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
25/06/04 18:35:24 [main]: WARN DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
Connected to: Apache Hive (version 3.1.3)
Driver: Hive JDBC (version 3.1.3)
Transaction isolation: TRANSACTION_REPEATABLE_READ
0: jdbc:hive2://> CREATE TABLE logs (
. . . . > user_a STRING,
. . . . > visit_time STRING,
. . . . > query STRING
. . . . >)
. . . . > ROW FORMAT DELIMITED
. . . . > FIELDS TERMINATED BY '\t';
25/06/04 18:35:29 [HiveServer2-Background-Pool: Thread-52]: ERROR metastore.RetryingHMSHandler: AlreadyEx



EJERCICIO 3 Identificar cuales usuarios visitan paginas mejores rankeadas en promedio

1. CREATE TABLE pages (...)

Hive crea una tabla administrada llamada pages con tres columnas:

- name de tipo STRING
- url de tipo STRING
- time de tipo STRING
- ROW FORMAT DELIMITED: Indica que es un formato de texto delimitado

- FIELDS TERMINATED BY ' ': Define el espacio como separador de campos
- Se crea la carpeta en HDFS: /user/hive/warehouse/pages
- Se registran los metadatos en el Metastore de Hive
- No se cargan datos aún, solo se define la estructura

2. LOAD DATA LOCAL INPATH ... INTO TABLE pages

```sql
LOAD DATA LOCAL INPATH '/home/hivehadoop/visits.log'

OVERWRITE INTO TABLE pages;
```

Carga datos desde un archivo local al sistema de archivos de HDFS y los asocia a la tabla.

- LOCAL INPATH: Indica que el archivo está en el sistema de archivos local
- OVERWRITE: Elimina cualquier dato existente en la tabla antes de cargar los nuevos
- Copia el archivo visits.log a HDFS en: /user/hive/warehouse/pages/
- Interpreta cada línea del archivo como un registro
- Separa los campos usando espacios (como se definió en CREATE TABLE)
- Asigna los valores a las columnas en orden: name, url, time

3. CREATE TABLE avg_visit AS SELECT ...

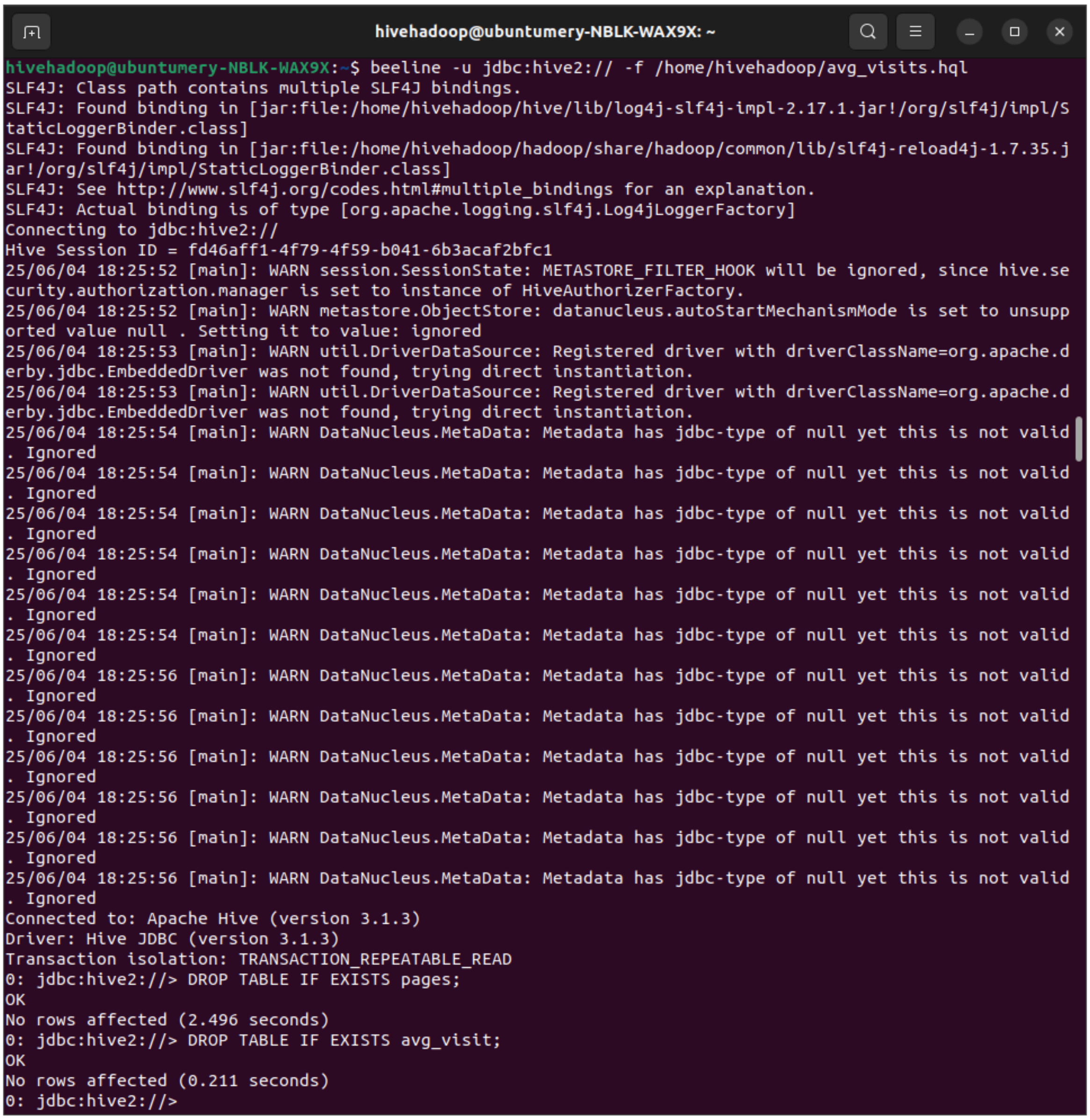
```sql
CREATE TABLE avg_visit AS

SELECT name, count(*) AS num_pages

FROM pages

GROUP BY name;
```

Crea una nueva tabla con estadísticas de visitas por nombre.

1. FROM pages: Lee todos los registros de la tabla pages
2. GROUP BY name: Agrupa los registros por el campo name
3. count(*) AS num_pages: Cuenta cuántos registros hay en cada grupo
4. CREATE TABLE avg_visit AS: Almacena el resultado en una nueva tabla física

La tabla avg_visit contendrá:

- Una columna name (los valores únicos de la tabla original)
- Una columna num_pages (el conteo de registros para cada nombre)



```
hivehadoop@ubuntumery-NBLK-WAX9X: ~
hivehadoop@ubuntumery-NBLK-WAX9X:~$ beeline -u jdbc:hive2:// -f /home/hivehadoop/avg_visits.hql
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/home/hivehadoop/hive/lib/log4j-slf4j-impl-2.17.1.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/hivehadoop/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.35.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Connecting to jdbc:hive2://
Hive Session ID = fd46aff1-4f79-4f59-b041-6b3acaf2bfc1
25/06/04 18:25:52 [main]: WARN session.SessionState: METASTORE_FILTER_HOOK will be ignored, since hive.security.authorization.manager is set to instance of HiveAuthorizerFactory.
25/06/04 18:25:52 [main]: WARN metastore.ObjectStore: datanucleus.autoStartMechanismMode is set to unsupported value null . Setting it to value: ignored
25/06/04 18:25:53 [main]: WARN util.DriverDataSource: Registered driver with driverClassName=org.apache.derby.jdbc.EmbeddedDriver was not found, trying direct instantiation.
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25/06/04 18:25:54 [main]: WARN DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
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25/06/04 18:25:56 [main]: WARN DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
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Connected to: Apache Hive (version 3.1.3)
Driver: Hive JDBC (version 3.1.3)
Transaction isolation: TRANSACTION_REPEATABLE_READ
0: jdbc:hive2://> DROP TABLE IF EXISTS pages;
OK
No rows affected (2.496 seconds)
0: jdbc:hive2://> DROP TABLE IF EXISTS avg_visit;
OK
No rows affected (0.211 seconds)
0: jdbc:hive2://>
```
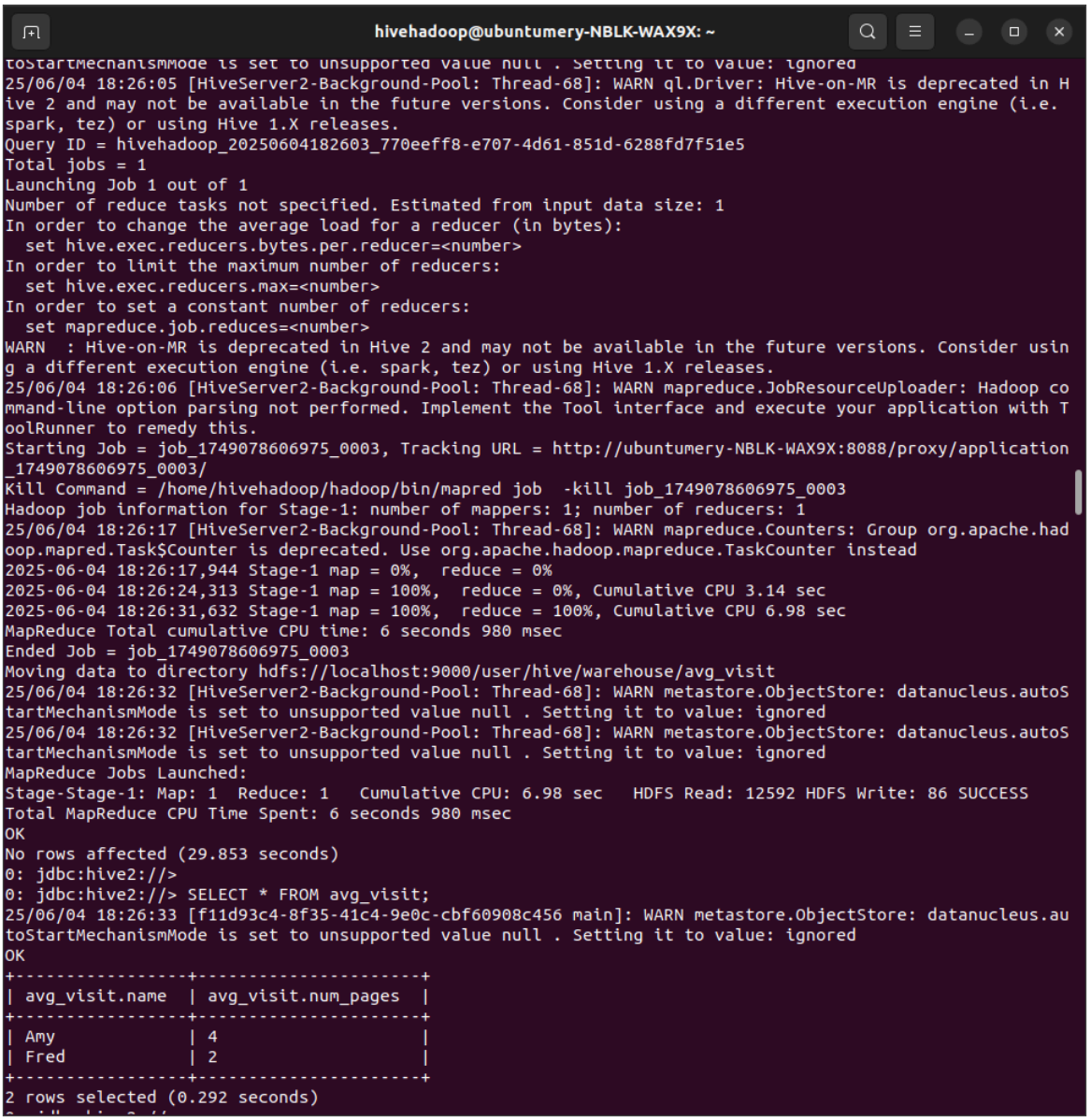



EJERCICIO 4: Identificar cuáles usuarios visitan paginas mejores rankeadas en promedio

SQL

```sql
SET hive.auto.convert.join = false;

DROP TABLE IF EXISTS visits;

CREATE TABLE visits (
```

```
  name STRING,
  url STRING,
  `time` STRING
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ' ';
LOAD DATA LOCAL INPATH '/home/hivehadoop/visits.log'
OVERWRITE INTO TABLE visits;
DROP TABLE IF EXISTS pages;
CREATE TABLE pages (
  url STRING,
  pagerank DECIMAL(3,2)
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ' ';
LOAD DATA LOCAL INPATH '/home/hivehadoop/pages.log'
OVERWRITE INTO TABLE pages;
DROP TABLE IF EXISTS rank_result;
CREATE TABLE rank_result AS
SELECT pr.name
FROM (
  SELECT v.name, AVG(p.pagerank) AS prank
  FROM visits v
```

```
  JOIN pages p ON (v.url = p.url)
  GROUP BY v.name
) pr
WHERE pr.prank > 0.5;
```

1. SET hive.auto.convert.join = false;

- Esta línea desactiva la conversión automática de JOINs a MapJoins (una optimización interna).
- Esto asegura que los JOINs se ejecuten como reparticiones completas, útil para evitar errores si las tablas son grandes.

2. DROP TABLE IF EXISTS visits;

```
SQL
CREATE TABLE visits (
 name STRING,
 url STRING,
 `time` STRING
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ' ';
```

Elimina la tabla visits si ya existe, para evitar errores por duplicación.

- Crea la tabla visits con los siguientes campos:
  - name: nombre del usuario
  - url: página que visitó
  - time: fecha/hora de la visita (el campo se pone entre comillas por ser palabra reservada)
- ROW FORMAT DELIMITED: indica que los campos están separados por un delimitador.
- FIELDS TERMINATED BY ' ': el delimitador es un espacio. Se crea en HDFS: /user/hive/warehouse/visits

3. LOAD DATA LOCAL INPATH '/home/hivehadoop/visits.log' ...

- Carga los datos del archivo visits.log en la tabla visits.
- LOCAL indica que el archivo está en el sistema local (no en HDFS).
- OVERWRITE reemplaza cualquier dato anterior.

Copia visits.log a la carpeta de la tabla en HDFS y lo parsea usando espacio como delimitador.

4. DROP TABLE IF EXISTS pages;

```
SQL

CREATE TABLE pages (
 url STRING,
 pagerank DECIMAL(3,2)
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ' ';
```

- Elimina la tabla pages si existe.
- Crea la tabla pages con:
  - url: URL de la página
  - pagerank: ranking de importancia de la página (máximo 9.99)

Se crea /user/hive/warehouse/pages

5. LOAD DATA LOCAL INPATH '/home/hivehadoop/pages.log' ...

- Carga los datos del archivo pages.log en la tabla pages.
- Los campos se separan por espacios.
- Se espera que el archivo tenga líneas como: http://x.com 0.85

---

6. DROP TABLE IF EXISTS rank_result;

```
SQL

CREATE TABLE rank_result AS
```

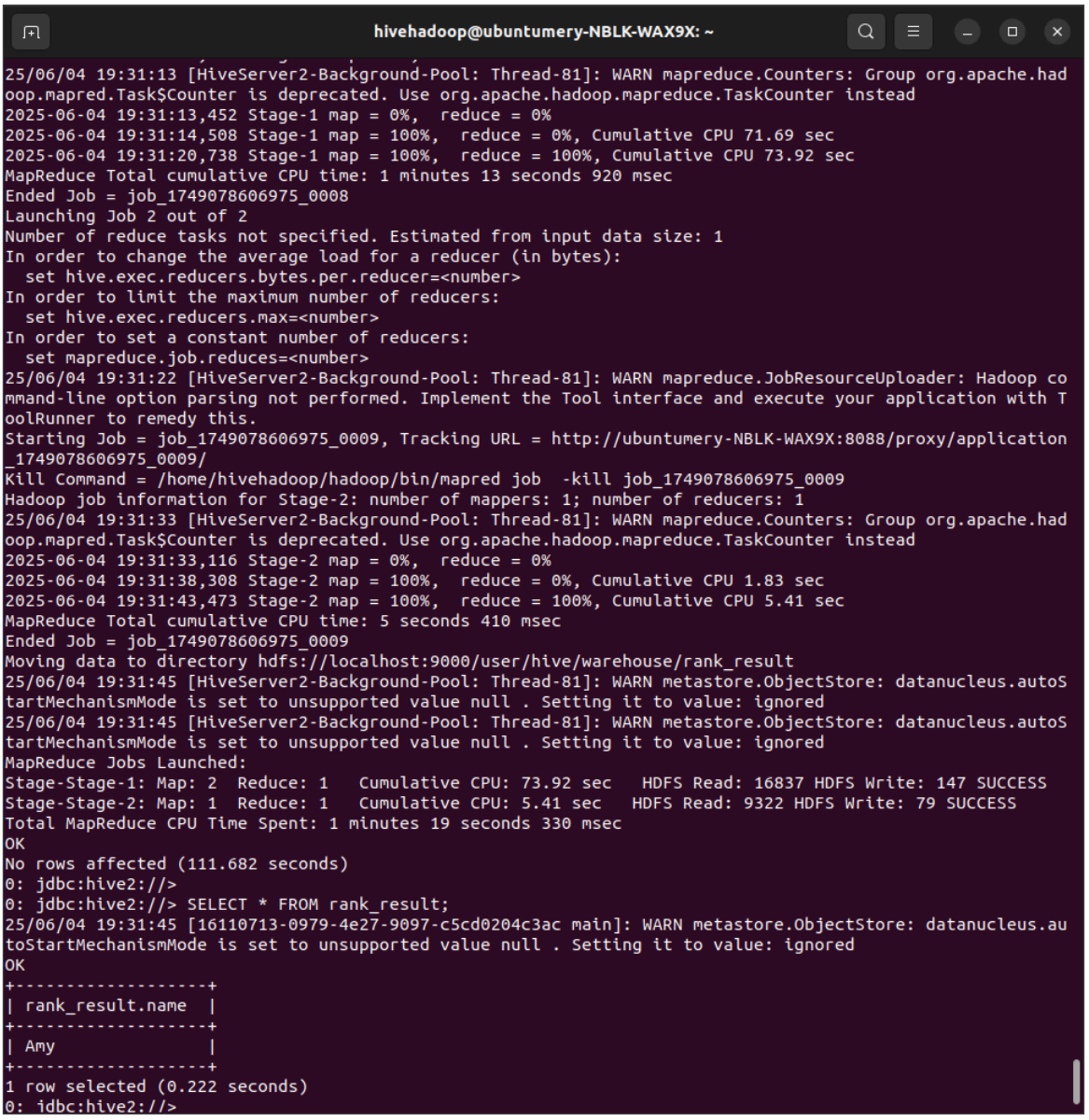
```
SELECT pr.name
FROM (
  SELECT v.name, AVG(p.pagerank) AS prank
  FROM visits v
  JOIN pages p ON (v.url = p.url)
  GROUP BY v.name
) pr
WHERE pr.prank > 0.5;
```

- Elimina la tabla rank_result si existe.
- Crea la tabla rank_result con los nombres de usuarios cuyo promedio de PageRank es mayor a 0.5.

1. JOIN entre visits y pages usando la URL como clave.
2. Agrupa por nombre de usuario (v.name) y calcula el promedio de PageRank (AVG(p.pagerank)).
3. Filtra solo los usuarios con promedio mayor a 0.5.
4. Crea una nueva tabla con los resultados.

hivehadoop@ubuntumery-NBLK-WAX9X: ~

```
hivehadoop@ubuntumery-NBLK-WAX9X:~$ beeline -u jdbc:hive2:// -f /home/hivehadoop/rank.hql
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/home/hivehadoop/hive/lib/log4j-slf4j-impl-2.17.1.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/hivehadoop/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.35.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Connecting to jdbc:hive2://
Hive Session ID = aa3526a3-0b05-483a-be54-6aeb4cb7f4f1
25/06/04 19:29:45 [main]: WARN session.SessionState: METASTORE_FILTER_HOOK will be ignored, since hive.security.authorization.manager is set to instance of HiveAuthorizerFactory.
25/06/04 19:29:45 [main]: WARN metastore.ObjectStore: datanucleus.autoStartMechanismMode is set to unsupported value null . Setting it to value: ignored
25/06/04 19:29:46 [main]: WARN util.DriverDataSource: Registered driver with driverClassName=org.apache.derby.jdbc.EmbeddedDriver was not found, trying direct instantiation.
25/06/04 19:29:46 [main]: WARN util.DriverDataSource: Registered driver with driverClassName=org.apache.derby.jdbc.EmbeddedDriver was not found, trying direct instantiation.
25/06/04 19:29:47 [main]: WARN DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
25/06/04 19:29:47 [main]: WARN DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
25/06/04 19:29:47 [main]: WARN DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
25/06/04 19:29:47 [main]: WARN DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
25/06/04 19:29:47 [main]: WARN DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
25/06/04 19:29:47 [main]: WARN DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
25/06/04 19:29:48 [main]: WARN DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
25/06/04 19:29:48 [main]: WARN DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
25/06/04 19:29:48 [main]: WARN DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
25/06/04 19:29:48 [main]: WARN DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
25/06/04 19:29:48 [main]: WARN DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
25/06/04 19:29:48 [main]: WARN DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
Connected to: Apache Hive (version 3.1.3)
Driver: Hive JDBC (version 3.1.3)
Transaction isolation: TRANSACTION_REPEATABLE_READ
0: jdbc:hive2://> SET hive.auto.convert.join = false;
No rows affected (0.067 seconds)
0: jdbc:hive2://>
0: jdbc:hive2://> DROP TABLE IF EXISTS visits;
OK
No rows affected (1.915 seconds)
0: jdbc:hive2://> CREATE TABLE visits (
```


hivehadoop@ubuntumery-NBLK-WAX9X: ~
25/06/04 19:31:13 [HiveServer2-Background-Pool: Thread-81]: WARN mapreduce.Counters: Group org.apache.hadoop.mapred.Task$Counter is deprecated. Use org.apache.hadoop.mapreduce.TaskCounter instead
2025-06-04 19:31:13,452 Stage-1 map = 0%, reduce = 0%
2025-06-04 19:31:14,508 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 71.69 sec
2025-06-04 19:31:20,738 Stage-1 map = 100%, reduce = 100%, Cumulative CPU 73.92 sec
MapReduce Total cumulative CPU time: 1 minutes 13 seconds 920 msec
Ended Job = job_1749078606975_0008
Launching Job 2 out of 2
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
25/06/04 19:31:22 [HiveServer2-Background-Pool: Thread-81]: WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
Starting Job = job_1749078606975_0009, Tracking URL = http://ubuntumery-NBLK-WAX9X:8088/proxy/application_1749078606975_0009/
Kill Command = /home/hivehadoop/hadoop/bin/mapred job -kill job_1749078606975_0009
Hadoop job information for Stage-2: number of mappers: 1; number of reducers: 1
25/06/04 19:31:33 [HiveServer2-Background-Pool: Thread-81]: WARN mapreduce.Counters: Group org.apache.hadoop.mapred.Task$Counter is deprecated. Use org.apache.hadoop.mapreduce.TaskCounter instead
2025-06-04 19:31:33,116 Stage-2 map = 0%, reduce = 0%
2025-06-04 19:31:38,308 Stage-2 map = 100%, reduce = 0%, Cumulative CPU 1.83 sec
2025-06-04 19:31:43,473 Stage-2 map = 100%, reduce = 100%, Cumulative CPU 5.41 sec
MapReduce Total cumulative CPU time: 5 seconds 410 msec
Ended Job = job_1749078606975_0009
Moving data to directory hdfs://localhost:9000/user/hive/warehouse/rank_result
25/06/04 19:31:45 [HiveServer2-Background-Pool: Thread-81]: WARN metastore.ObjectStore: datanucleus.autoStartMechanismMode is set to unsupported value null . Setting it to value: ignored
25/06/04 19:31:45 [HiveServer2-Background-Pool: Thread-81]: WARN metastore.ObjectStore: datanucleus.autoStartMechanismMode is set to unsupported value null . Setting it to value: ignored
MapReduce Jobs Launched:
Stage-Stage-1: Map: 2 Reduce: 1   Cumulative CPU: 73.92 sec   HDFS Read: 16837 HDFS Write: 147 SUCCESS
Stage-Stage-2: Map: 1 Reduce: 1   Cumulative CPU: 5.41 sec   HDFS Read: 9322 HDFS Write: 79 SUCCESS
Total MapReduce CPU Time Spent: 1 minutes 19 seconds 330 msec
OK
No rows affected (111.682 seconds)
0: jdbc:hive2://>
0: jdbc:hive2://> SELECT * FROM rank_result;
25/06/04 19:31:45 [16110713-0979-4e27-9097-c5cd0204c3ac main]: WARN metastore.ObjectStore: datanucleus.autoStartMechanismMode is set to unsupported value null . Setting it to value: ignored
OK
+------------------+
| rank_result.name |
+------------------+
| Amy |
+------------------+
1 row selected (0.222 seconds)
0: jdbc:hive2://>